# Trustworthy Machine Learning

## Certified Adversarial Learning

**Sangdon Park**

POSTECH

# Motivation

- Heuristic adversarial learning often fails against powerful adversaries.

| | CIFAR10 | | | | | |
|---|---|---|---|---|---|---|
| | Simple | Wide | Simple | Wide | Simple | Wide |
| Natural | 92.7% | 95.2% | 87.4% | 90.3% | 79.4% | 87.3% |
| FGSM | 27.5% | 32.7% | 90.9% | 95.1% | 51.7% | 56.1% |
| PGD | 0.8% | 3.5% | 0.0% | 0.0% | 43.7% | 45.8% |
| | (a) Standard training | | (b) FGSM training | | (c) PGD training | |

  - FGSM training and FGSM attacks: $90.9\%$ accuracy :)
  - FGSM training and PGD attacks: $0.0\%$ accuracy :(

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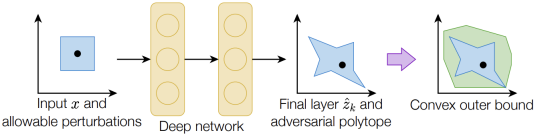
  - FGSM training and FGSM attacks: $90.9\%$ accuracy :)
  - FGSM training and PGD attacks: $0.0\%$ accuracy :(
- Can we learn a classifier robust to **any** small perturbations?

# Certified Adversarial Learning

- Convex outer approximation [Kolter and Wong, 2017]




✓ Certified!

$$\max_{\|\delta\|_\infty \leq \varepsilon} \ell(f, x+\delta, y) \leq U(\varepsilon, f, x, y)$$

✗ Not scalable :(

# Certified Adversarial Learning

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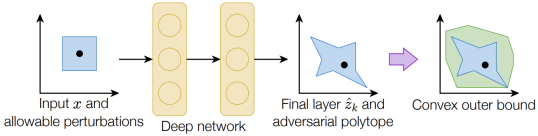
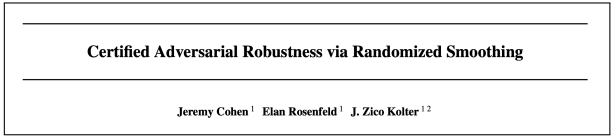



  - ✓ Certified!

$$\max_{\|\delta\|_\infty \leq \varepsilon} \ell(f, x+\delta, y) \leq U(\varepsilon, f, x, y)$$

  - ✗ Not scalable :(

- Randomized smoothing: a post-hoc method


**Certified Adversarial Robustness via Randomized Smoothing**

**Jeremy Cohen** [1] **Elan Rosenfeld** [1] **J. Zico Kolter** [1 2]

  - ✓ (Probably) Certified!
  - ✓ Scalable!

# A Goodness Definition: Robustness

**"Hard" Robustness**

$$\max_{\|\delta\|_p \leq \varepsilon} f(x+\delta) = f(x)$$

- $f : \mathcal{X} \to \mathcal{Y}$: a classifier

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**"Hard" Robustness**

$$\max_{\|\delta\|_p \leq \varepsilon} f(x+\delta) = f(x)$$

- $f : \mathcal{X} \to \mathcal{Y}$: a classifier
- The constraint on the perturbation $\delta$ can be more general.
- It does not matter whether $f(x)$ is correct.

# A Certified Method: Randomized Smoothing

**Smoothed Classifier**

$$g(x) := \arg\max_{c \in \mathcal{Y}} \mathbb{P}\left\{f(x+\delta) = c\right\} \quad \text{where} \quad \delta \sim \mathcal{N}(0, \sigma^2 I)$$

- $g : \mathcal{X} \to \mathcal{Y}$: a smoothed classifier
- $\sigma$ is related to the maximum perturbation $\varepsilon$.
- Easier than checking the "hard" robustness condition

# Robustness Guarantee

## Binary Classification

### Theorem

*Suppose that* $\underline{p_A} \in (0.5, 1]$ *satisfies*

$$\mathbb{P}\left\{f(x+\delta) = c_A\right\} \geq \underline{p_A} \quad \textit{where} \quad \delta \sim \mathcal{N}(0, \sigma^2 I).$$

*Then, we have* $g(x+\delta) = c_A$ *if*

$$\|\delta\|_2 < \sigma \Phi^{-1}(\underline{p_A}).$$

- $c_A$: the most probable class when $f$ classifies $x + \varepsilon$
- $p_A$: the chance that $f$ classifies $x + \delta$ by $c_A$
- $\underline{p_A}$: the lower bound of $p_A$
- $\Phi^{-1}$: the inverse of the standard Gaussian CDF

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- $\Phi^{-1}$: the inverse of the standard Gaussian CDF
- Here, we assume that we can compute $\underline{p_A}$.
- We can compute the *data-dependent* maximum perturbation to be robust!

# Robustness Guarantee: A Proof Sketch (1/3)

## Binary Classification

- Fix a perturbation $\delta$.
- From the definition of $g$, we have

$$
\begin{aligned}
g(x+\delta) &:= \arg\max_c \mathbb{P}\left\{f(x+\varepsilon+\delta)=c\right\} \quad \text{where} \quad \varepsilon \sim \mathcal{N}(0,\sigma^2 I) \\
&= \arg\max_c \mathbb{P}\left\{f(x+\varepsilon')=c\right\} \quad \text{where} \quad \varepsilon' \sim \mathcal{N}(\delta,\sigma^2 I) \\
&\overset{?}{=} c_A
\end{aligned}
\tag{1}
$$

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## Binary Classification

- Fix a perturbation $\delta$.
- From the definition of $g$, we have

$$\begin{aligned} g(x+\delta) &:= \arg\max_c \mathbb{P}\left\{f(x+\varepsilon+\delta)=c\right\} \quad \text{where} \quad \varepsilon \sim \mathcal{N}(0,\sigma^2 I) \\ &= \arg\max_c \mathbb{P}\left\{f(x+\varepsilon')=c\right\} \quad \text{where} \quad \varepsilon' \sim \mathcal{N}(\delta,\sigma^2 I) \\ &\overset{?}{=} c_A \end{aligned} \tag{1}$$

- We wish to prove (1) for any classifier $f$ under some condition. How?
  - $f$ can be any classifier, which is not easy to analyze.
  - Consider a surrogate classifier that bounds the probability and is easier to analyze, *e.g.*,

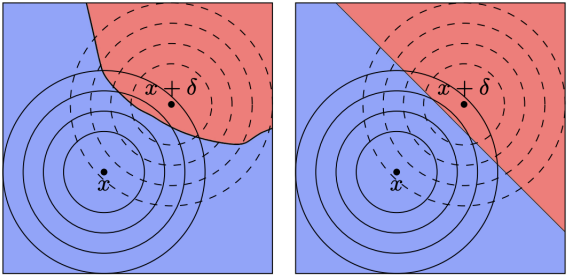
$$\mathbb{P}\left\{f(x+\varepsilon')=c_A\right\} \geq \min_{f':\mathbb{P}\{f'(x+\varepsilon)=c_A\}\geq \underline{p_A}} \mathbb{P}\left\{f'(x+\varepsilon')=c_A\right\} > \frac{1}{2} \quad \Longrightarrow \quad g(x+\delta)=c_A.$$

# Robustness Guarantee: A Proof Sketch (2/3)

## Binary Classification

- Interestingly, $f^*$ is linear (due to the Neyman-Perason lemma), where

$$f^* = \arg \min_{f' : \mathbb{P}\{f'(x+\varepsilon)=c_A\} \geq \underline{p_A}} \mathbb{P}\left\{f'(x+\varepsilon') = c_A\right\}$$




  - There could be a non-linear classifier but we can find a corresponding linear classifier with the same mininum value.

# Robustness Guarantee: A Proof Sketch (3/3)

## Binary Classification

- We have a closed-form solution of $f^*$:

$$f^*(x') := \begin{cases} c_A & \text{if } \delta^T(x' - x) \leq \sigma \|\delta\|_2 \Phi^{-1}(\underline{p_A}) \\ c_B & \text{otherwise} \end{cases}.$$

- This (non-trivially) implies the following mininum value:

$$\min_{f' : \mathbb{P}\{f'(x+\varepsilon)=c_A\} \geq \underline{p_A}} \mathbb{P}\left\{f'(x+\varepsilon') = c_A\right\} = \mathbb{P}\left\{f^*(x+\varepsilon') = c_A\right\} = \Phi\left(\Phi^{-1}(\underline{p_A}) - \frac{\|\delta\|_2}{\sigma}\right)$$

- The above probability should be larger than $\frac{1}{2}$, *i.e.*,

$$\Phi\left(\Phi^{-1}(\underline{p_A}) - \frac{\|\delta\|_2}{\sigma}\right) > \frac{1}{2} \quad \Longrightarrow \quad \|\delta\|_2 < \sigma \Phi^{-1}(\underline{p_A}).$$

# Robustness Guarantee

## Multi-class Classification

**Theorem**

*Suppose that* $\underline{p_A}, \overline{p_B} \in [0,1]$ *satisfies*

$$\mathbb{P}\left\{f(x+\varepsilon) = c_A\right\} \geq \underline{p_A} \geq \overline{p_B} \geq \max_{c \neq c_A} \mathbb{P}\left\{f(x+\varepsilon) = c\right\}.$$

*Then, we have* $g(x+\delta) = c_A$ *for all* $\|\delta\|_2 \leq R$*, where*

$$R := \frac{\sigma}{2}\left(\Phi^{-1}(\underline{p_A}) - \Phi^{-1}(\overline{p_B})\right).$$

- $c_A$: the most probable label (with probability at least $\underline{p_A}$)
- $c_B := \arg\max_{c \neq c_A} \mathbb{P}\left\{f(x+\varepsilon) = c\right\}$: the second-most probable label (with probability at most $\overline{p_B}$)

# Prediction

**function** PREDICT($f, \sigma, x, n, \alpha$)
  `counts` $\leftarrow$ SAMPLEUNDERNOISE($f, x, n, \sigma$)
  $\hat{c}_A, \hat{c}_B \leftarrow$ top two indices in `counts`
  $n_A, n_B \leftarrow$ `counts`$[\hat{c}_A]$, `counts`$[\hat{c}_B]$
  **if** BINOMPVALUE($n_A, n_A + n_B, 0.5$) $\leq \alpha$ **return** $\hat{c}_A$
  **else return** ABSTAIN

- Recall the randomized smoothing method:

$$g(x) := \arg\max_{c \in \mathcal{Y}} \mathbb{P}\left\{f(x+\delta) = c\right\} \quad \text{where} \quad \delta \sim \mathcal{N}(0, \sigma^2 I)$$

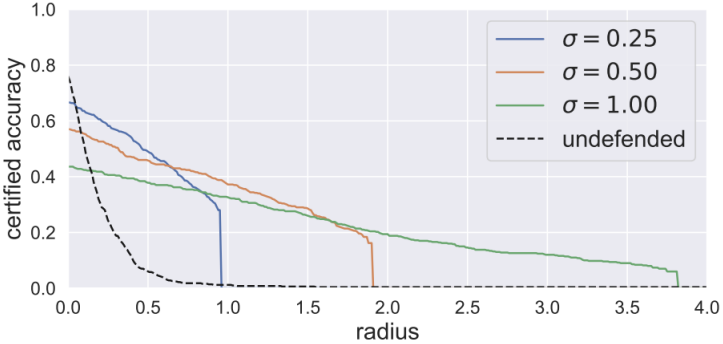
  1. Draw $n$ noisy perturbations $\delta_1, \ldots, \delta_n$.
  2. Empirically compute the most probable and the second most probably labels, *i.e.,* $\hat{c}_A$ and $\hat{c}_B$.
  3. If $\hat{c}_A$ is drawn from the binomial distribution with $p = 0.5$, return $\hat{c}_A$.
- Alternatively, you can use the (our-favorite) binomial tail bound.

# Certification in Evaluation

*# certify the robustness of $g$ around $x$*
**function** CERTIFY($f, \sigma, x, n_0, n, \alpha$)
  `counts0` $\leftarrow$ SAMPLEUNDERNOISE($f, x, n_0, \sigma$)
  $\hat{c}_A \leftarrow$ top index in `counts0`
  `counts` $\leftarrow$ SAMPLEUNDERNOISE($f, x, n, \sigma$)
  $\underline{p_A} \leftarrow$ LOWERCONFBOUND(`counts`$[\hat{c}_A], n, 1-\alpha$)
  **if** $\underline{p_A} > \frac{1}{2}$ **return** prediction $\hat{c}_A$ and radius $\sigma\,\Phi^{-1}(\underline{p_A})$
  **else return** ABSTAIN

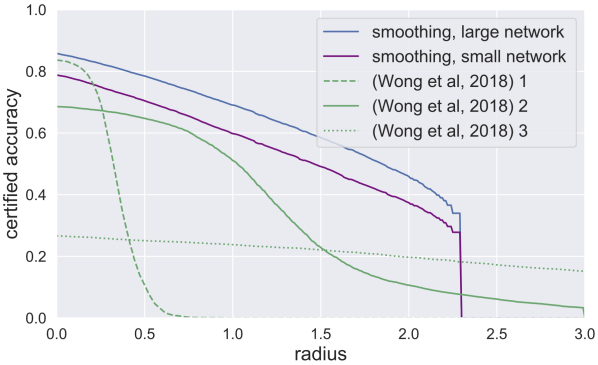
1. Compute $\underline{p_A}$ via the binomial tail bound.
2. Compute the robust radius, *i.e.*, $\sigma\Phi^{-1}(\underline{p_A})$.
3. If (a desired radius) $\leq \sigma\Phi^{-1}(\underline{p_A})$, then "certified".

# Results: ImageNet




- Classifier: ResNet-50
- undefended: a classifier with heuristic adversarial training (using $\ell_2$ adversarial attacks)
- perturbation: $\|\delta\|_2 \leq$ (radius) $=$ (maximum perturbation size)

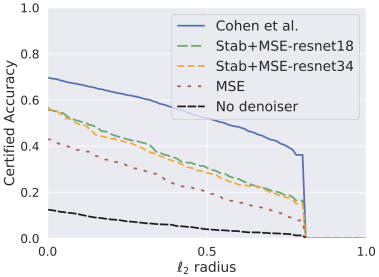
# Results: Comparison




- (maybe) on MNIST
- Baseline: deterministic robustness guarantee
- randomized smoothing: high-probability guarantee

# Limitation of Randomized Smoothing

- Randomized smoothing requires retraining (*e.g.*, Gaussian data augmentation).




  - Cohen et al.: Randomized smoothing with retraining
  - No denoiser: Randomized smoothing without retraining
- How to avoid retraining?

# Denoise Gaussian Noise

**Denoised Smoothing:**
**A Provable Defense for Pretrained Classifiers**

**Hadi Salman**
hasalman@microsoft.com
Microsoft Research

**Mingjie Sun**
mingjies@cs.cmu.edu
CMU

**Greg Yang**
gragyang@microsoft.com
Microsoft Research

**Ashish Kapoor**
akapoor@microsoft.com
Microsoft Research

**J. Zico Kolter**
zkolter@cs.cmu.edu
CMU

- A classifier randomized smoothing needs to be robust to Gaussian noise for better certified robustness.
- How about using denoised smoothing and then use the randomized smoothing?

# Denoised Smoothing

**Randomized Smoothing:**

$$g(x) := \arg\max_{c \in \mathcal{Y}} \mathbb{P}\left\{f(x+\delta) = c\right\} \quad \text{where} \quad \delta \sim \mathcal{N}(0, \sigma^2 I)$$

- Applicable for any classifier $f$

**Denoised Smoothing:**

$$g(x) := \arg\max_{c \in \mathcal{Y}} \mathbb{P}\left\{f(\mathcal{D}(x+\delta)) = c\right\} \quad \text{where} \quad \delta \sim \mathcal{N}(0, \sigma^2 I)$$

- $\mathcal{D} : \mathcal{X} \to \mathcal{X}$: a denoiser that hopefully removes $\delta$.
- Consider a NEW classifier $f \circ \mathcal{D}$ and then enjoy randomized smoothing.
- Retraining $f$ is not required.

# How to Train a Denoiser?

**MSE objective:**

$$L_{\mathsf{MSE}} := \mathop{\mathbb{E}}_{x,y,\delta} \|\mathcal{D}(x+\delta) - x\|_2^2$$

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$$L_{\text{MSE}} := \mathop{\mathbb{E}}_{x,y,\delta} \|\mathcal{D}(x+\delta) - x\|_2^2$$

✗ Does not consider the accuracy of a classifier.

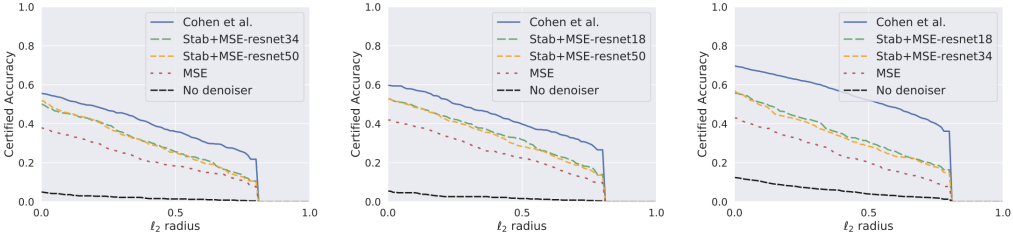
**Stability objective:**

$$L_{\text{Stab}} := \mathop{\mathbb{E}}_{x,y,\delta} \ell_{CE}(F(\mathcal{D}(x+\delta)), f(x)) \quad \text{where} \quad \delta \sim \mathcal{N}(0, \sigma^2 I)$$

- $f : \mathcal{X} \to \mathcal{Y}$: a hard classifier
- $F : \mathcal{X} \to [0,1]^{|\mathcal{Y}|}$: a soft classifer

✓ Find a denoiser $Ds$ that does not change predictions by the classifier $f$.

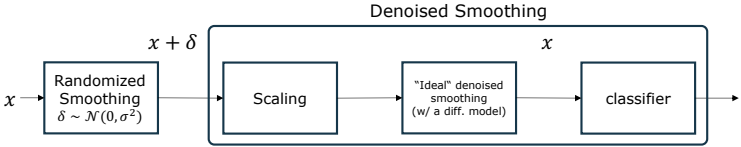
# Results




(a) ResNet-18 (b) ResNet-34 (c) ResNet-50

- The denoised smoothing without retraining is quite similar to the randomized smoothing with retraining.
- But not outperform the retraining one. How can we train a better denoiser?

# Diffusion Models as Denoisers




**Assumptions**:

- A diffusion model assumes the following noise model:

$$x_t := \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \cdot \mathcal{N}(0, \mathbf{I}),$$

  where $x_0$ is an initial example, $t$ is a timestep, and $\alpha_t$ is any noise scheduler (monotonically decreasing in $t$).
  - Under this mose model, an ideal denoiser finds $x_0$ from $x_t$.

**Method:**

- Find $t^*$ for randomized smoothing that fits to the noise model for a diffusion model, *i.e.,*

$$\text{find} \quad t \quad \text{subj. to} \quad x_0 + \mathcal{N}(0, \sigma^2 \mathbf{I}) \approx \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \cdot \mathcal{N}(0, \mathbf{I})$$

# Conclusion

- Randomized smoothing provides a simple defense mechanism.
- Desnoised smoothing does not require to retrain a classifier (but still requires training the denoiser).
- Recently, the denoised smoothing was improved via denoising diffusion models [Carlini et al., 2023].

| | | | Certified Accuracy at $\varepsilon$ (%) | | | | |
|---|---|---|---|---|---|---|---|
| Method | Off-the-shelf | Extra data | 0.5 | 1.0 | 1.5 | 2.0 | 3.0 |
| PixelDP (Lecuyer et al., 2019) | ○ | ✗ | $^{(33.0)}$16.0 | - | - | | |
| RS (Cohen et al., 2019) | ○ | ✗ | $^{(67.0)}$49.0 | $^{(57.0)}$37.0 | $^{(57.0)}$29.0 | $^{(44.0)}$19.0 | $^{(44.0)}$12.0 |
| SmoothAdv (Salman et al., 2019) | ○ | ✗ | $^{(65.0)}$56.0 | $^{(54.0)}$43.0 | $^{(54.0)}$37.0 | $^{(40.0)}$27.0 | $^{(40.0)}$20.0 |
| Consistency (Jeong & Shin, 2020) | ○ | ✗ | $^{(55.0)}$50.0 | $^{(55.0)}$44.0 | $^{(55.0)}$34.0 | $^{(41.0)}$24.0 | $^{(41.0)}$17.0 |
| MACER (Zhai et al., 2020) | ○ | ✗ | $^{(68.0)}$57.0 | $^{(64.0)}$43.0 | $^{(64.0)}$31.0 | $^{(48.0)}$25.0 | $^{(48.0)}$14.0 |
| Boosting (Horváth et al., 2022a) | ○ | ✗ | $^{(65.6)}$57.0 | $^{(57.0)}$44.6 | $^{(57.0)}$**38.4** | $^{(44.6)}$**28.6** | $^{(38.6)}$**21.2** |
| DRT (Yang et al., 2021) | ○ | ✗ | $^{(52.2)}$46.8 | $^{(55.2)}$44.4 | $^{(49.8)}$**39.8** | $^{(49.8)}$**30.4** | $^{(49.8)}$**23.4** |
| SmoothMix (Jeong et al., 2021) | ○ | ✗ | $^{(55.0)}$50.0 | $^{(55.0)}$43.0 | $^{(55.0)}$**38.0** | $^{(40.0)}$26.0 | $^{(40.0)}$20.0 |
| ACES (Horváth et al., 2022b) | ◐ | ✗ | $^{(63.8)}$54.0 | $^{(57.2)}$42.2 | $^{(55.6)}$35.6 | $^{(39.8)}$25.6 | $^{(44.0)}$19.8 |
| Denoised (Salman et al., 2020) | ◐ | ✗ | $^{(60.0)}$33.0 | $^{(38.0)}$14.0 | $^{(38.0)}$6.0 | - | - |
| Lee (Lee, 2021) | ● | ✗ | 41.0 | 24.0 | 11.0 | - | - |
| **Ours** | ● | ✓ | $^{(82.8)}$**71.1** | $^{(77.1)}$**54.3** | $^{(77.1)}$**38.1** | $^{(60.0)}$**29.5** | $^{(60.0)}$13.1 |

# Reference I

N. Carlini, F. Tramer, K. D. Dvijotham, L. Rice, M. Sun, and J. Z. Kolter. (certified!!) adversarial robustness for free!, 2023.

J. Z. Kolter and E. Wong. Provable defenses against adversarial examples via the convex outer adversarial polytope. *arXiv preprint arXiv:1711.00851*, 2017.